Electrical and Mechanical Portfolio

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1. AURC 2026 Competition

The manufacturing process of the entire rocket spanned the whole three-week mid-lecture break, where it continued for the next six weeks. In total, this build took a process of nine weeks, excluding the work needed to be done during the mid-year break.

Originally, this timeline was set for the rocket to be ready for test flight during the mid-year break in retrospect for the AURC 2026 competition. However, logistical issues resulted in the team not having any of the specified motors needed in time for the competition due to exportation rules. As each rocket motor has a set dimension, this meant that the rocket design would be incompatible with motors of differing diameters. This resulted in the team pulling out this year and instead launching in New Zealand, with hopes of returning next year.

Although the outcome was unfortunate without knowing the results we would have had, I learned so many skills from this project. It was an amazing experience getting to see firsthand how a competition rocket is built from scratch and personally having an impact on the outcome onboard a passionate team of students. A collection of photos across the design process is shown below.

1.1 Boat Tail

The carbon fibre sections were cut out at 0° and 45° angles, which I alternated on the weaves onto a 3D-printed mould that had three layers of mould release. Combining different fibre angles makes the boat tail strong in every direction. This part is epoxied using wet layup techniques and West System 105 Epoxy Resin and 206 Hardener. It is then layered with perforated film and external peel ply to separate it from the vacuum seal. This step is necessary to push the epoxy along the edges. To reduce repetition for other parts of the rocket, assume that infusion and wet layup techniques follow the same steps above. A pressure of < 20 mBar is the aim for all vacuum-sealed parts. Figure 1 shows this in the final step and the untrimmed version. Extra epoxy is brushed on the inside where dry patches were found.

FIG. 1: VACUUM SEALED BOAT TAIL AND FINISHED BOAT TAIL.

1.2 Fins

Four fin plates were needed for the design of the rocket. The plate lay-up had fifteen layers of carbon fibre with alternating layers at 0° and 45° angles. An approximate thickness of 3.5 mm was preferred to provide the stiffness and rigidity for its appropriate use. The sheets were cut into 340 × 180 mm sections to have a buffer in case they were cut too small for the fins. Mould release was applied to a glass plate, and acetone was used to clean the edges for back tape to be put on for the vacuum bag. Figure 2 shows the vacuum-sealed plate lay-up, the cut version of the fin plate, and the raw carbon fibre.

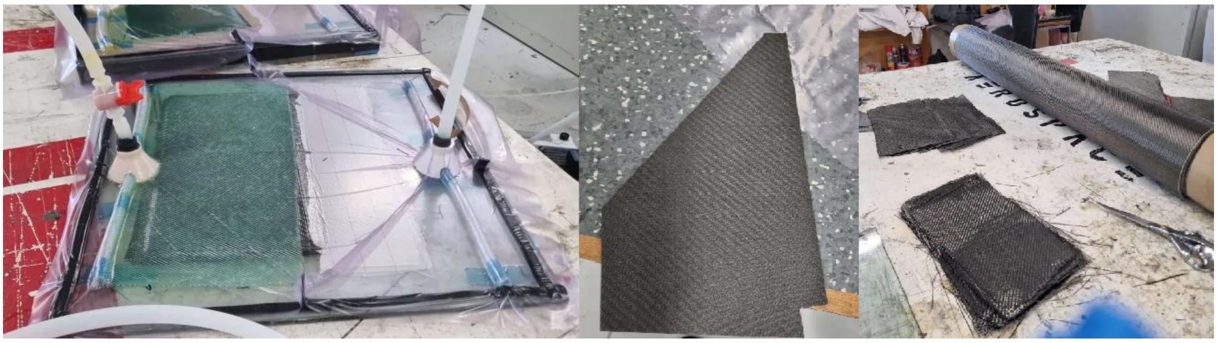

FIG. 2: VACUUM SEALED FIN PLATE, WATERJET CUT FIN, AND RAW CARBON FIBRE PROCESSING.

In the beginning of the project, one of the finished fin plates was six layers thick, which caused it to have a bit of flex to it. This meant it was not stiff enough, which is why every fin plate from then on was redone to have fifteen layers. The result created a product which was much stiffer and had no bend to it. The thickness of the fin plates varies due to epoxy resin being absorbed, but it averaged to around 260 g for both epoxy and carbon fibre for fifteen layers.

In the end, a total of seven fin plates were created, which occurred due to a number of factors. Weak vacuuming caused by older machines happened as the Kirkwood workshop is shared among UC Motorsport and UC Human Power clubs. The stronger vacuum is shared and, at times, is not available. This is important as we want the epoxy to fully saturate from the top to the bottom of the plate. Another reason may have also been the placement of the infusion mesh. Figure 3 demonstrates where the infusion mesh should lay compared to the mesh covering the whole carbon fibre squares. There wasn't enough margin on one side of the plate lay-up, and so it did not create the pressure drop for the epoxy to fully saturate towards the bottom of the plate. Changing the steps when curing the carbon-fibre allowed for a stronger product.

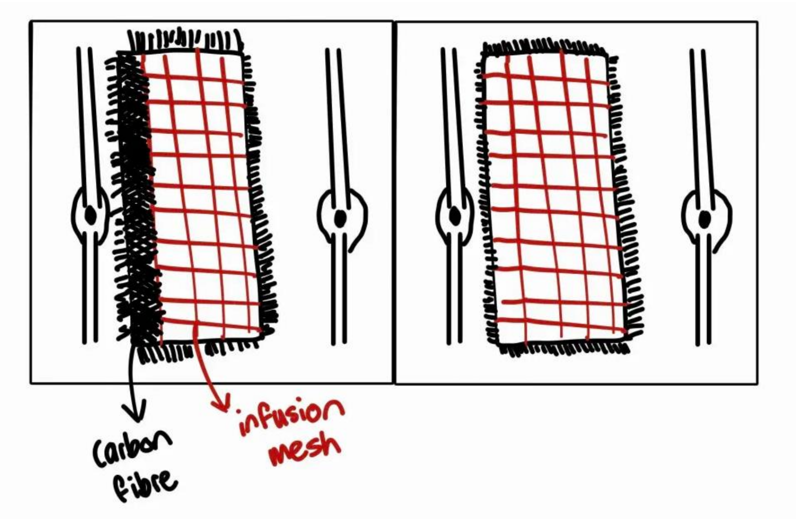

FIG. 3: SKETCH OF END PROCESS AND INITIAL PROCESS RESPECTIVELY.

These were mitigated by using the University machines in the composites lab, which created satisfactory fin plates. Figure 4 shows the fin plates that did not cure properly and so were not used.

FIG. 4: FIN PLATES THAT CURED IMPROPERLY.

1.3 Air Brakes and Avionics Sled

As the whole design of the rocket is derived from the 2024 Spaceport design, old hardware such as the air brakes were also integrated into the booster and payload tube. A problem was found when looking at the holes of the air brakes. The dimensioning of the holes was not equispaced and had differing angles for each of the holes, so they were not consistent. When measured, a 17.4° offset was found on the payload side and about 22.5° for the booster side. These holes needed to match the holes of the carbon fibre tube, and it was much harder to create holes from the inside out. Having no CAD files to go off and not having the original files also made this harder. Figure 5 shows the payload and booster-side air brakes.

FIG. 5: PAYLOAD AND BOOSTER SIDE AIR BRAKES RESPECTIVELY.

Due to the air brakes being a very expensive component of the rocket ($3–4k), and the lack of funding to source a new one, two solutions were thought out. The easiest solution was to drill an extra set of holes that had consistent dimensioning across the whole part. For clarification, Figure 6 is a sketch of this idea, which shows the cross marks as the holes that would be drilled. It was agreed upon that this would not have a substantial effect on the structural integrity of the part, so it was deemed a viable option. A solution I proposed was to create a jig that fits over the air brakes and carbon fibre tube, which maps out the original holes onto the tube and can then be drilled out.

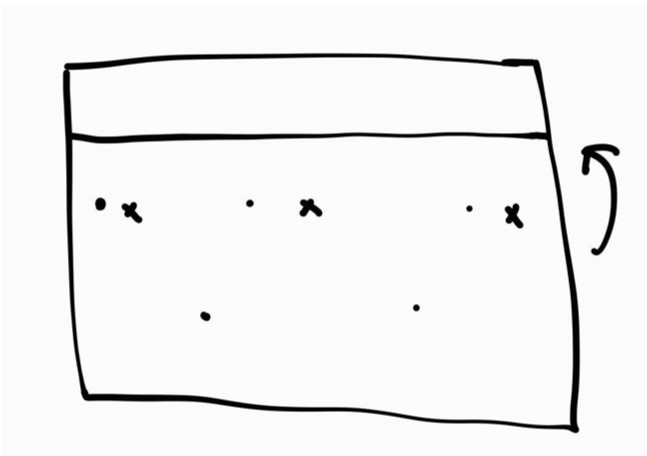

FIG. 6: SKETCH OF THE FRONT VIEW PAYLOAD AIRBRAKES OF PROPOSED IDEA 1

As the lead wanted the airbrakes to maintain its original appearance, the jig solution was preferred. I started this by using a piece of paper that is wrapped along the outside of the air brakes. Holes were then poked through using a pen. This is used for rough measurements which can then be used for the jig. Creating two of these jigs - one for the payload and booster side tubes - was used to drill holes into the tubes. Figure 7 shows the jigs being 3D printed.

FIG. 7: PAYLOAD AND BOOSTER JIGS BEING 3D PRINTED.

The first iteration tested the alignment of the airbrakes. This needed minor tweaks to the vertical spacing between the two sets of holes on the booster side jig. Figure 8 shows the second iterations of the booster side jigs fitting over the air brakes.

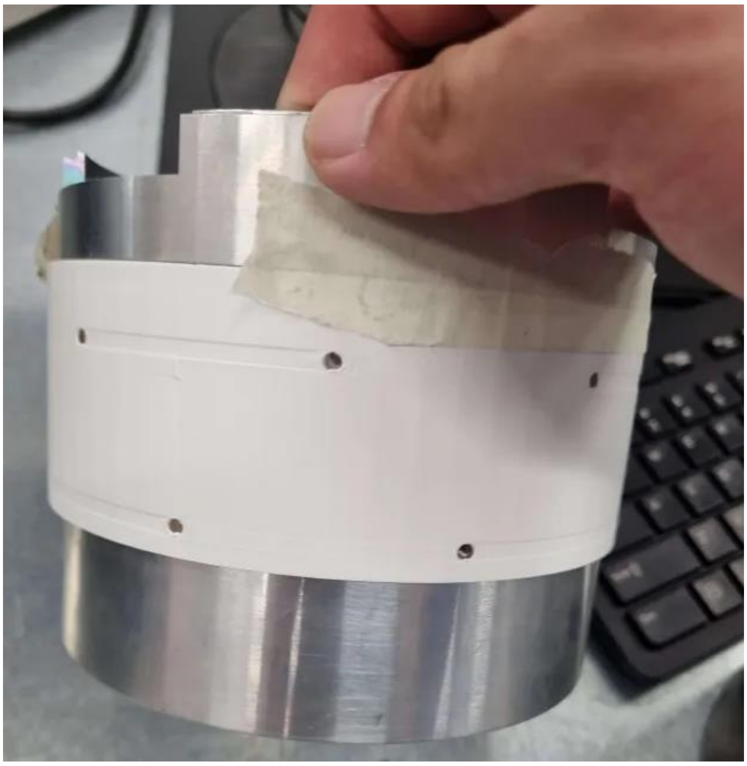

FIG. 8: BOOSTER SIDE JIG OVER THE AIR BRAKES.

Using these jigs for the carbon fibre tubes allowed for a template to be drilled out. Cumulative error was built up by minor shaking from drilling and the rough measurements taken from the jig to the air brakes. This meant that some holes needed to be filed in order to fit the M3-sized screws properly. This process took hours of filing and refitting screws to double-check the fit of the jig holes and the air brakes. The screws were then countersunk for aerodynamics so they sat flush against the tube. Figure 9 shows the payload-side tube drilled with holes using the jig and screwed in with the air brakes.

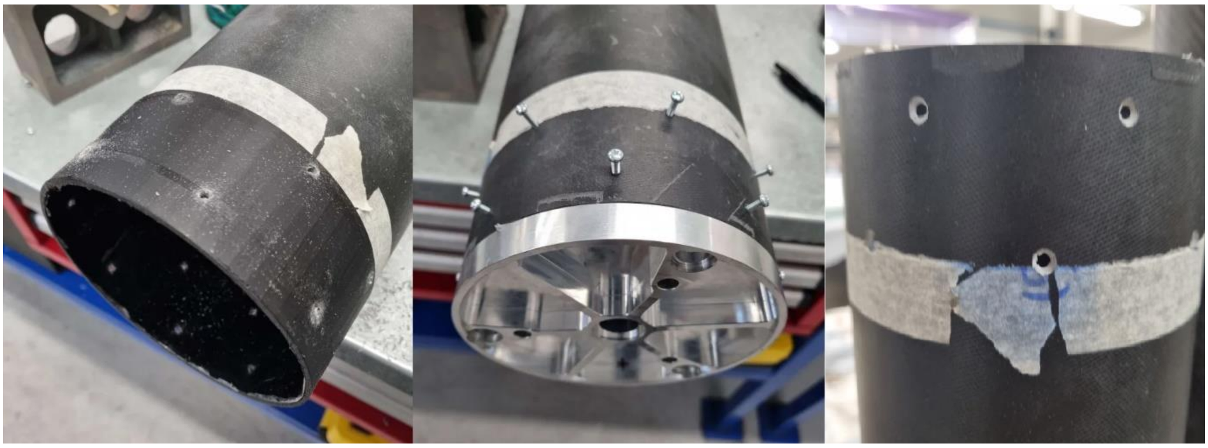

FIG. 9: PAYLOAD JIG OVER THE TUBE, AIR BRAKE ASSEMBLY, AND COUNTERSUNK HOLES

This process was repeated for the booster side air brakes and the booster tube. Figure 10 shows the lower air frame assembly with the payload and booster attached together along with the fin cut outs.

Unfortunately, around this time my phone became unrecoverable. This meant the loss of photos I collected for the remainder of this project was void!

1.4 Nosecone

The nosecone mould was originally going to be made from MDF as seen from Spaceport 2024. However, a 3D printed mould was used instead as an alternative due to technicians being unavailable at the time of this project. The mould is printed out of three separate parts and glued together using cyanoacrylate. A thick plaster called bog is spread on top to fill in the crevices of the surface and sanded in order to achieve a smooth finish in the final product. Eight layers of mould release is applied. Figure 11 shows the nosecone mould being made with the bogged 3D printed mould bogged up!

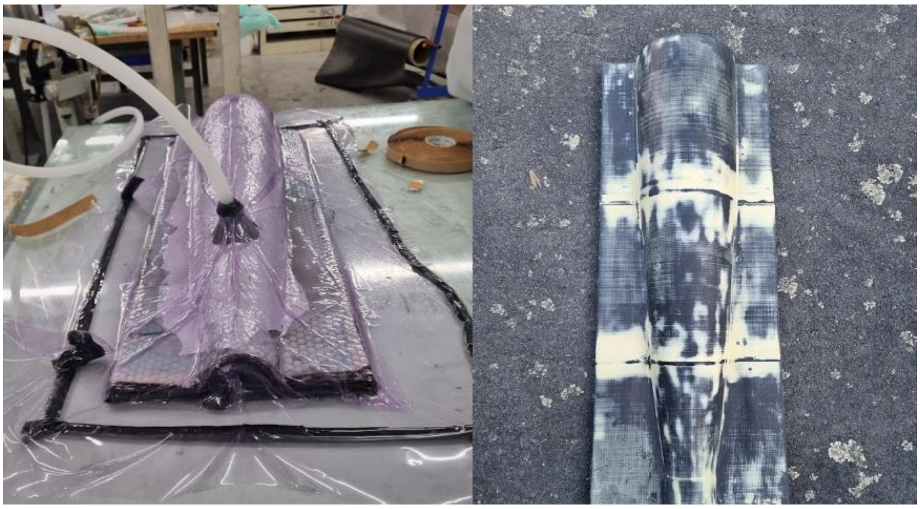

FIG. 11: NOSECONE MOULD LAYUP AND 3D PRINTED MOULD.

This was a necessary step as we needed the mould to cure an actual composite nosecone that would be used for the rocket. This needed two tries as the resin used was fast curing, which meant that the 3D print experienced too much heat in a short amount of time. This warped the PLA where Figure 12 shows the notable dent in the mould.

FIG. 12: AFTERMATH OF NOSECONE LAYUP.

A setback for the project due to the failed mould meant that a new 3D print had to be made. Rebogging and sanding the new mould with a new change using a slow-curing allowed for the 3D print to survive the lay-up process. Figure 13 shows the second attempt outcome with bog that is sanded down for a smooth finish. Note that both nosecones were made of the same materials being foam-core, glass fibre, fibreglass.

FIG. 13: SURVIVING BOGGED NOSECONE.

This nosecone mould allowed for the nosecone to be cured

1.5 Tip-to-tip

Sixty layers of prepreg are cut out for the individual fins. Four layers of twill carbon fibre are cut out for each side of the fins which become the final outer layer due to their nice finish. Eight layers of prepreg are applied to each side of the fin, consisting of different sizes according to the cutting stencils.

We start from the smallest layer to the largest because we want the fins to taper out with four leading-edge cut-outs of prepreg splayed over the edges. The vacuum bag used dimensions of

16 cm back tape, spaced 27.5 cm apart, so four sections of back tape were needed to match the number of fins. This is an important step because we want the composite to cure properly. Although it is possible to let it cure without vacuum bagging, we wanted to go out of our way to do a good job for this project. Four layers of external peel ply, already infused with resin were also added to provide a nice finish. Perforated blue film and breather were then added as the final layers before sealing the vacuum bag.

Originally, vacuum bagging the fins to the inside of the booster tube could not hold a strong vacuum (350 mbar). Ideally, the goal is to reach the lowest possible pressure reading, so redoing the vacuum bagging process the following day using the inside of the body tube was the solution. This allowed us to achieve a much better seal and reduce the pressure reading by a factor of seven, from 350 mbar down to 50 mbar. Figure 14 shows the tip to tip bagged up to cure with the 350mbar vacuum.

FIG. 14: VACUUM BAGGED TIP-TO-TIP.

Figure 15 shows version 1 and 2 respectively which version 2 will be used in the avionics sled for the project.

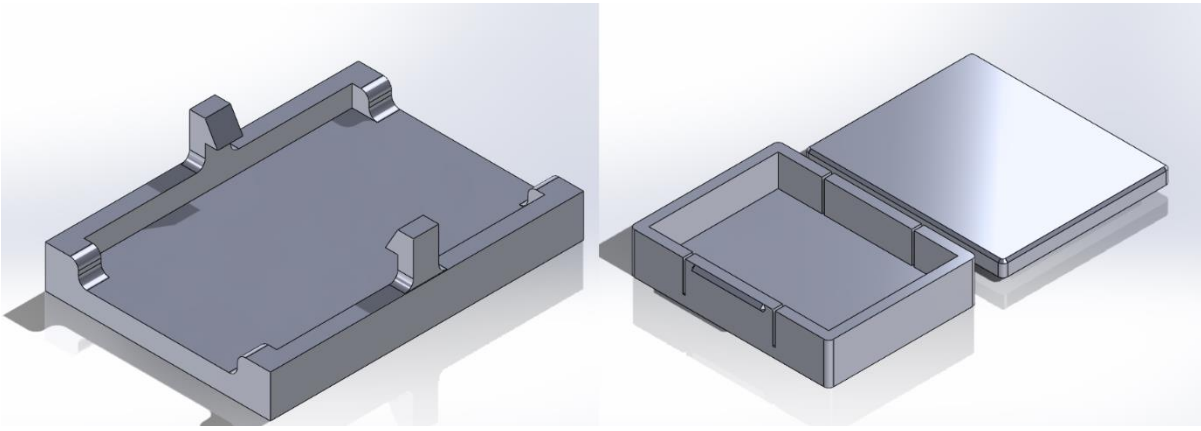

FIG. 15: FIRST AND SECOND VERSION OF BATTERY HOLDERS RESPECTIVELY.

2. UC Rocketry Project

The goal of this project was to build up a model kit of a quarter size scaled patriot missile using mechanical processes such as epoxy and sanding to end up with a finished product. The team consisted of two other members where I managed deadlines of how our rocket was progressing and fostering a learning environment. The whole timeline consisted of seven months to deliver a functioning rocket along with a report.

I personally did not document this project quite well in my first year, however, as a returning flight certification mentor, I am able to show more steps of this process. The figure shown below is the finished project with my team of friends!